

Santosh Sapkota

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TECHNICAL SKILLS

- **Programming:** Python
- **Machine Learning & Deep Learning:** Classical ML, CNN, RNN, Transfer Learning, Time-Series Modeling
- **Computer Vision & OCR:** OpenCV, Tesseract OCR, Image Preprocessing
- **Frameworks & Libraries:** PyTorch/TensorFlow, Scikit-learn, Pandas, NumPy, Matplotlib
- **Backend & MLOps:** FastAPI, Docker, Git, CI/CD, MLflow, DVC, Airflow
- **GenAI & NLP:** LLM-based Systems, RAG, LangChain, Multilingual Translation Pipelines

EXPERIENCE

Machine Learning Engineer

Jan 2026 – Present

Ishani Technology

Lalitpur, Nepal

Document Management System

- Built an end-to-end OCR pipeline for document digitization, combining computer vision preprocessing with Tesseract to handle noisy, scanned municipal documents.
- Designed a hybrid extraction system leveraging direct parsing for born-digital files and OCR fallback for scanned inputs, achieving near 100% accuracy on digital PDFs and ~85% on scanned printed documents.
- Integrated an LLM-based multilingual translation pipeline with parallel processing for efficient multi-page document conversion.
- Extended the document pipeline to handle attendance sheets, using OCR to extract class/student metadata and a separate CNN classifier (trained on EMNIST dataset) to detect handwritten Present/Absent achieving ~90% accuracy.

Machine Learning Intern

Aug 2025 – Sep 2025

Elevo Tech, Egypt

Remote

- Developed and evaluated end-to-end machine learning pipelines across 4 projects, utilizing SMOTE for class imbalance and performing rigorous feature engineering.
- Designed and trained a CNN-based image classification model using TensorFlow, implementing reproducible experimentation workflows aligned with MLOps practices.
- Automated data preprocessing steps, reducing manual data cleaning time by 40%.

PROJECTS

ICU Patient Deterioration Early-Warning System

Python, Airflow, Docker, FastAPI

MLOps & Pipeline Orchestration

[Code] [Demo]

- Architected a production-grade ML system to forecast ICU patient deterioration 6 hours in advance, achieving 90%+ recall.
- Operationalized the training lifecycle with MLflow, DVC, and Airflow.
- Deployed a containerized FastAPI microservice with CI/CD automation and implemented production monitoring using Prometheus, Grafana, and EvidentlyAI for drift detection.

AI Expense Intelligence System

FastAPI, OCR, RAG, Pinecone

Document Intelligence System

[Code] [Demo]

- Built an AI-powered document pipeline to extract structured expense data from PDFs and receipt images using OCR, LLM-based extraction, and confidence-based review.
- Implemented a RAG assistant combining SQL metadata filtering with Pinecone vector search for natural-language queries over approved financial documents.

Interpretable Breast Cancer Detection

TensorFlow, CNN

Medical Imaging

[Code] [Demo]

- Developed a deep learning pipeline for histopathology image classification using a custom CNN architecture.
- Implemented Grad-CAM to visualize model decision heatmaps, adding interpretability to "black box" predictions..
- Productionized the model via a FastAPI backend and Streamlit frontend, fully containerized and deployed on Render.

EDUCATION

Cosmos College of Management & Technology

Sep 2018 – Aug 2023

Bachelor of Computer Engineering

Lalitpur, Nepal

PUBLICATIONS

Sapkota, S. (2025). *Sapkota, S. (2025). Breast Cancer Diagnosis Using Custom CNN and Grad-CAM: An Interpretable Deep Learning Approach on Histopathology Images* Zenodo. doi:10.5281/zenodo.16900046